Sampling of analog signals

I.1. Analog and Digital Signals

Signals

- Signal = a measurable quantity which varies in time, space or some other variable
- Examples:
  - a voltage which varies in time (1D voltage signal)
  - sound pressure which varies in time (sound signal)
  - intensity of light which varies across a photo (2D image)
- Represented as a mathematical function, e.g. $v(t)$.

Glossary

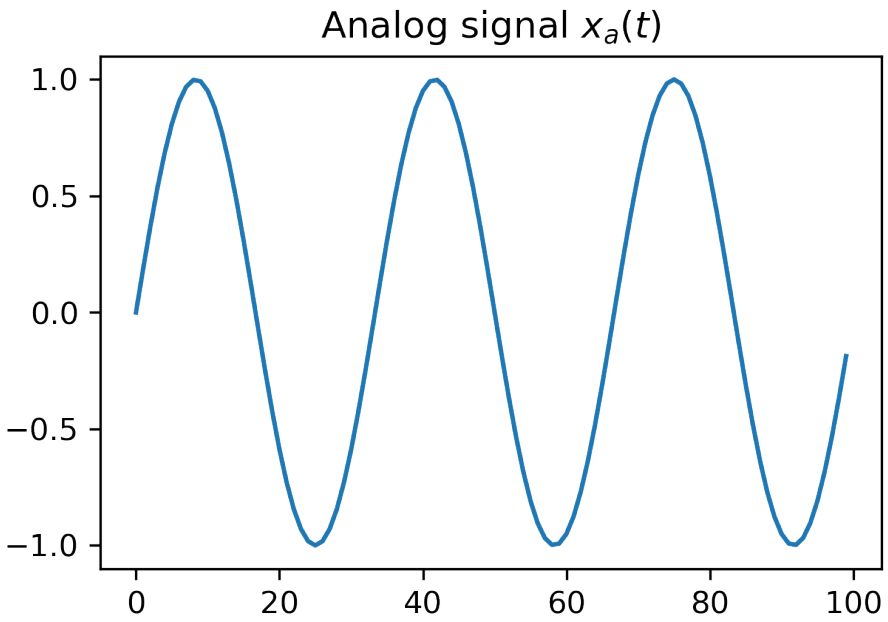
- Glossary:
  - "e.g." = "*exempli gratia*" (lat.) = "for example" (eng.) = "de exemplu" (rom.)
  - "i.e." = "*id est*" (lat) = "that is" (eng.) = "adică" (rom.)

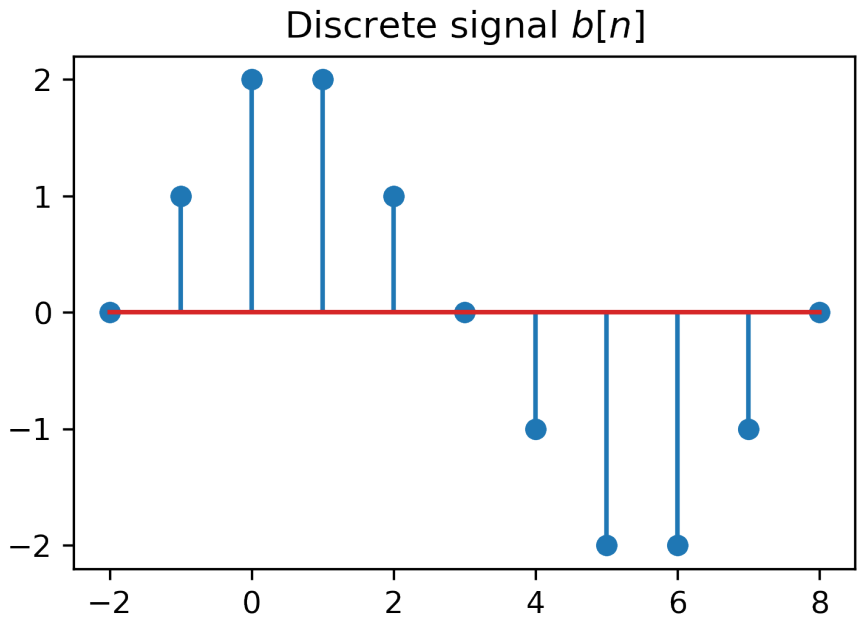
Signal dimension

- **Unidimensional** (1D) signal = a function of a single variable
  - Example: a voltage signal $v(t)$ only varies in time.
- **Multidimensional** (2D, 3D ... M-D) signal = a function of multiple variables
  - Example: intensity of a grayscale image $I(x, y)$ across the surface of a photo
- In these lectures we consider only 1D signals, but the theory is similar

Continuous and discrete signals

- Continuous (analog) signal = function of a continuous variable
  - Signal has a value for every possible value of the variable in the defined range
  - The variable may be defined only in a certain range (e.g. $t \in [0, 100]$), but it is a continuous range
- Discrete signal = function of a discrete variable
  - Signal has values only at certain discrete values (*samples*)
  - Indexed with integer numbers: $x[-1]$, $x[0]$, $x[1]$ etc.
  - Outside the samples, the signal is **not defined**





Notation

- We use the following notation:
- Continuous signal
  - Has **round parentheses**, e.g. $x_a(t)$
  - Sometimes has the a subscript
  - The variable is usually t (time)
  - $x_a(2.3) =$ the value of the signal $x_a(t)$ at $t = 2.3$
- Discrete signal
  - Has **square brackets**, e.g. $x[n]$
  - The variables are denoted as n or k (suggest natural numbers)
  - $x[3] =$ the value of the signal $x[n]$ for $n = 3$
  - $x[1.5] =$ does not exist

Continuous, sampled and digital signals

- **Analog signal**: continuous time, continuous amplitude
  - e.g. a microphone voltage $x_a(t)$
- **Sampled signal**: discrete time, continuous amplitude
  - sampling selects the values $x_a(nT_s)$
- **Digital signal**: discrete time, discrete amplitude
  - quantization rounds every sample to one of a finite number of levels
- **Sampling discretizes time; quantization discretizes amplitude**

Periodic signals

- A signal is **periodic** if the values repeat themselves after a certain time (**fundamental period**)
- Continuous signals:
  - Periodic: $x_a(t) = x_a(t + T), \forall t, \text{ where } T > 0$
  - T is usually measured in seconds (or some other unit)
- Discrete signals:
  - Periodic: $x[n] = x[n + N], \forall n, \text{ where } N > 0$
  - N **has no unit**, because it is just a number

Discrete frequency

- Harmonic signals have a frequency f:

$$x(t) = 2 \cdot \cos(2\pi \cdot 400 \cdot t + \frac{\pi}{3})$$

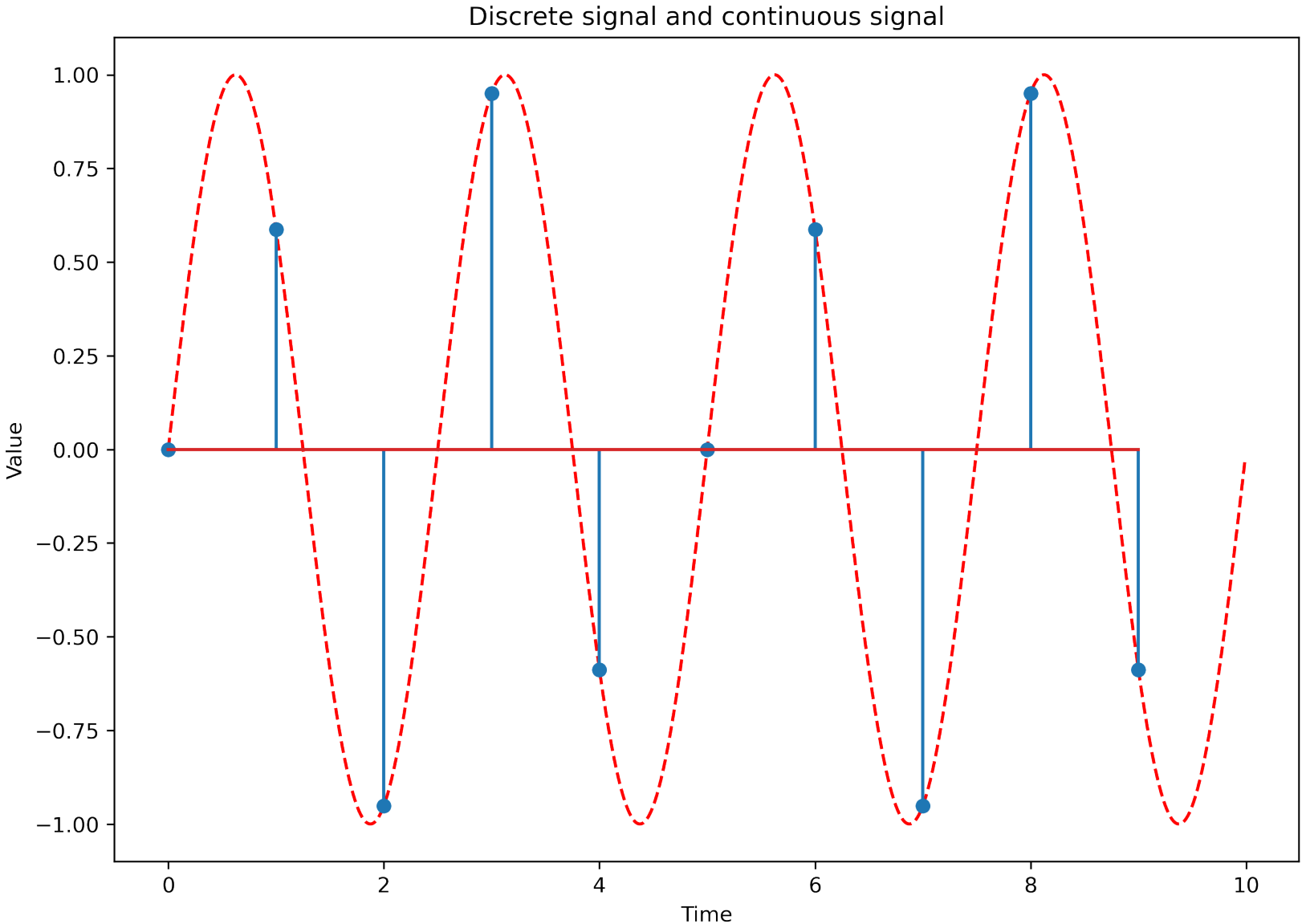
$$x[n] = 5 \cdot \sin(2\pi \cdot 0.12 \cdot n + \frac{\pi}{2})$$

- Pulsation $\omega = 2\ \pi \cdot$ frequency
- Continuous signals:
  - $T = \frac{1}{f}$ is usually measured in seconds (or some other unit)
  - $F = \frac{1}{T}$ is measured in Hz $= \frac{1}{s}$ (Hertz)
- Discrete signals:
  - N **has no unit**, because it is just a number
  - f **has no unit** also

Periodicity of a discrete sinusoid

- Consider
$$x[n] = \cos(2\pi f n)$$
- It is periodic **if and only if** f is rational
- If $f = \frac{p}{q}$ is in lowest terms, its fundamental period is
$$N = q$$
- Example: $f = \frac{3}{8}$ gives $N = 8$
- Contrast: $f = \frac{\sqrt{2}}{10}$ is irrational, so no finite period exists

Period and frequency



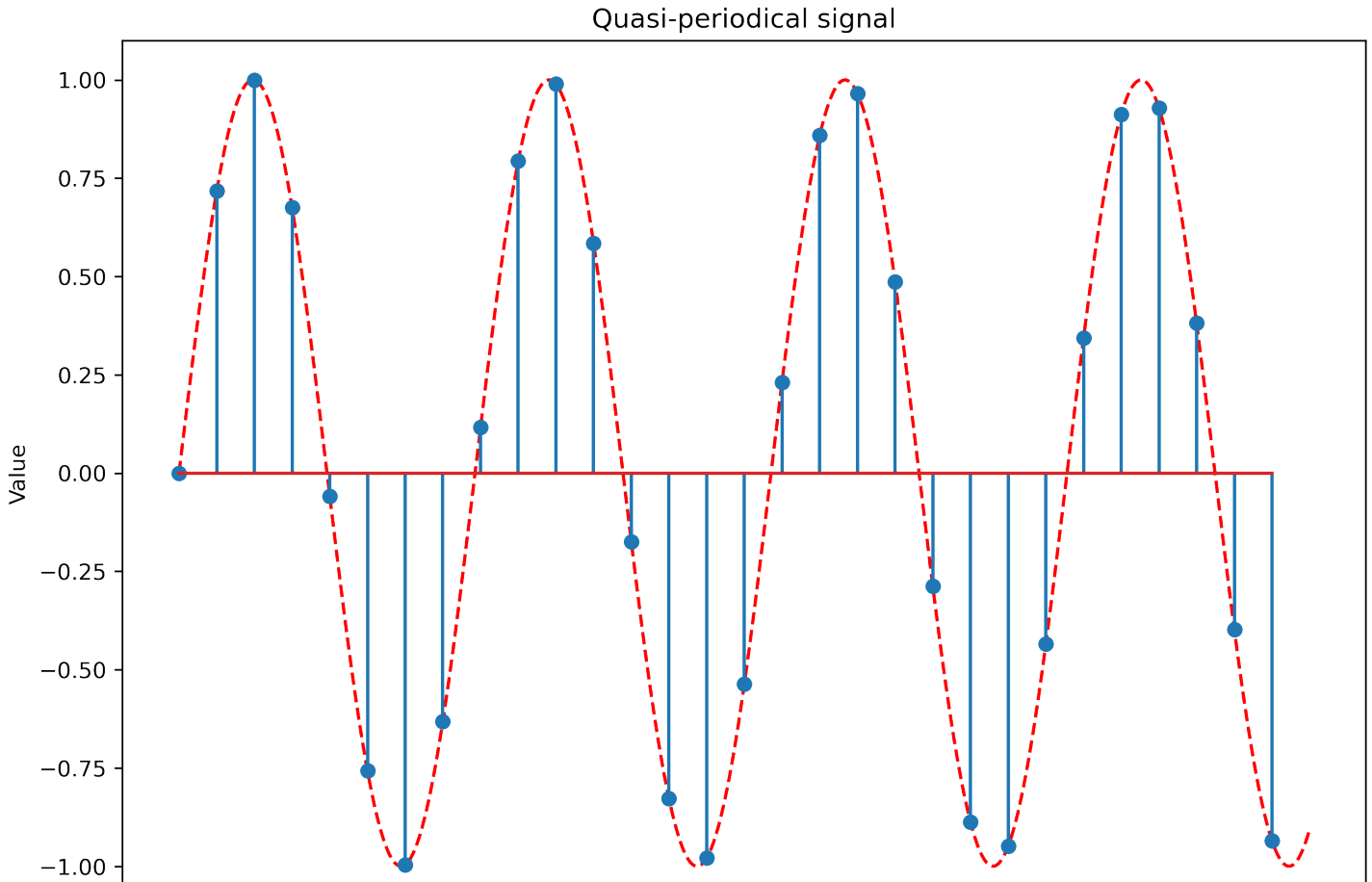
- Discrete signal period $N = 5$, frequency $f = 0.4$

Period and frequency

- Quasi-periodic signals:

$$x[n] = \sin(0.8 \cdot n)$$

- Frequency = irrational number, period = does not exist, but the signal is "almost periodic"



Checkpoint: periodicity

- For $x[n] = \cos(2\pi f n)$, decide whether it is periodic:
  - $f = \frac{5}{12}$
  - $f = \frac{\sqrt{3}}{8}$
- **Answer:** the first signal is periodic with $N = 12$; the second is not periodic
- Reducing $f = \frac{p}{q}$ to lowest terms is essential

Domain of definition

- **Finite-length** discrete signals $x[n]$:
  - have only a certain number N of samples (e.g. for n = 0, 1, ... N-1)
  - they are not defined outside these samples
  - can be represented as a **vector** of numbers (e.g. like in Matlab, C)
- **Infinite-length** discrete signals $x[n]$:
  - e.g. defined for $n = ... -2, -1, 0, 1, 2, ...$

Vector space of signals

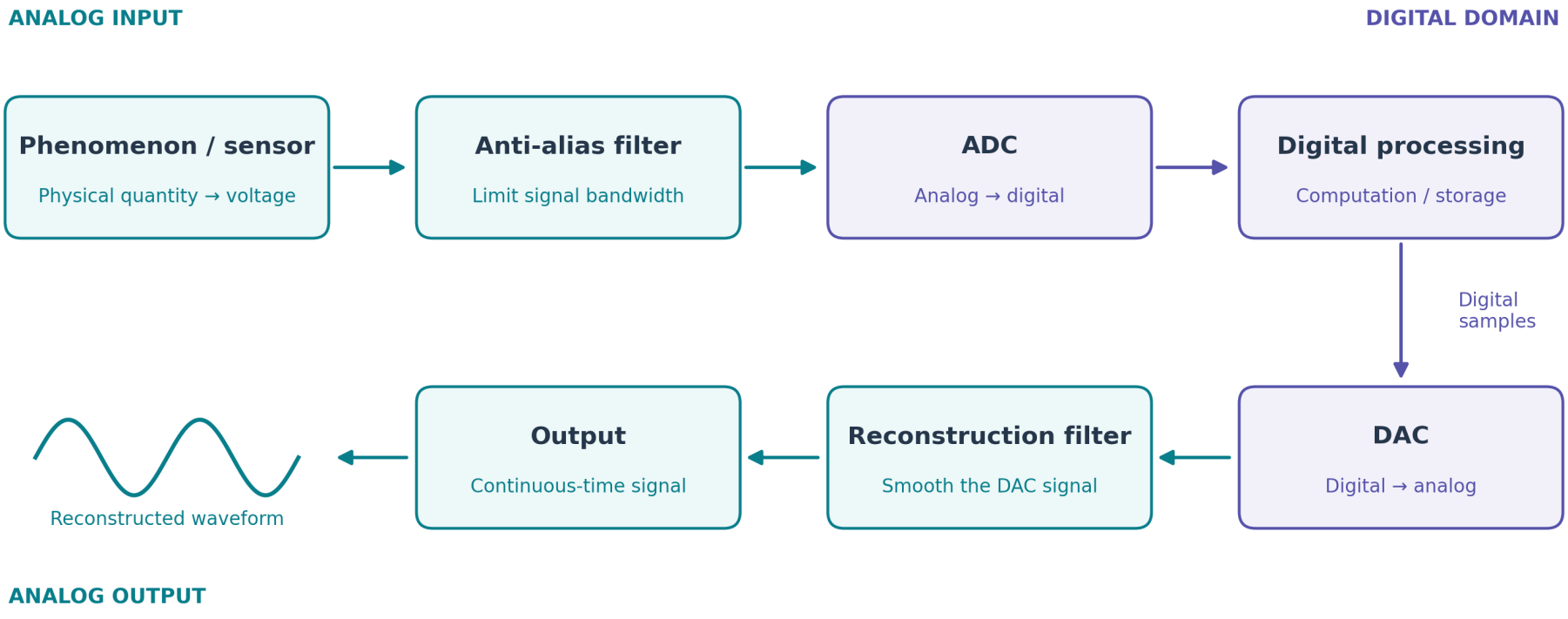
- All signals of a certain length N form a **vector space**
- In mathematics, a vector space $=$ a set V of elements $\{v\}$ (called "vectors' ') such that:
  - the sum of any two elements from V is still a member of V
  - any vector from V multiplied by a constant is still a member of V
- These properties can easily be verified for signals

I.2. Sampling

Sampling

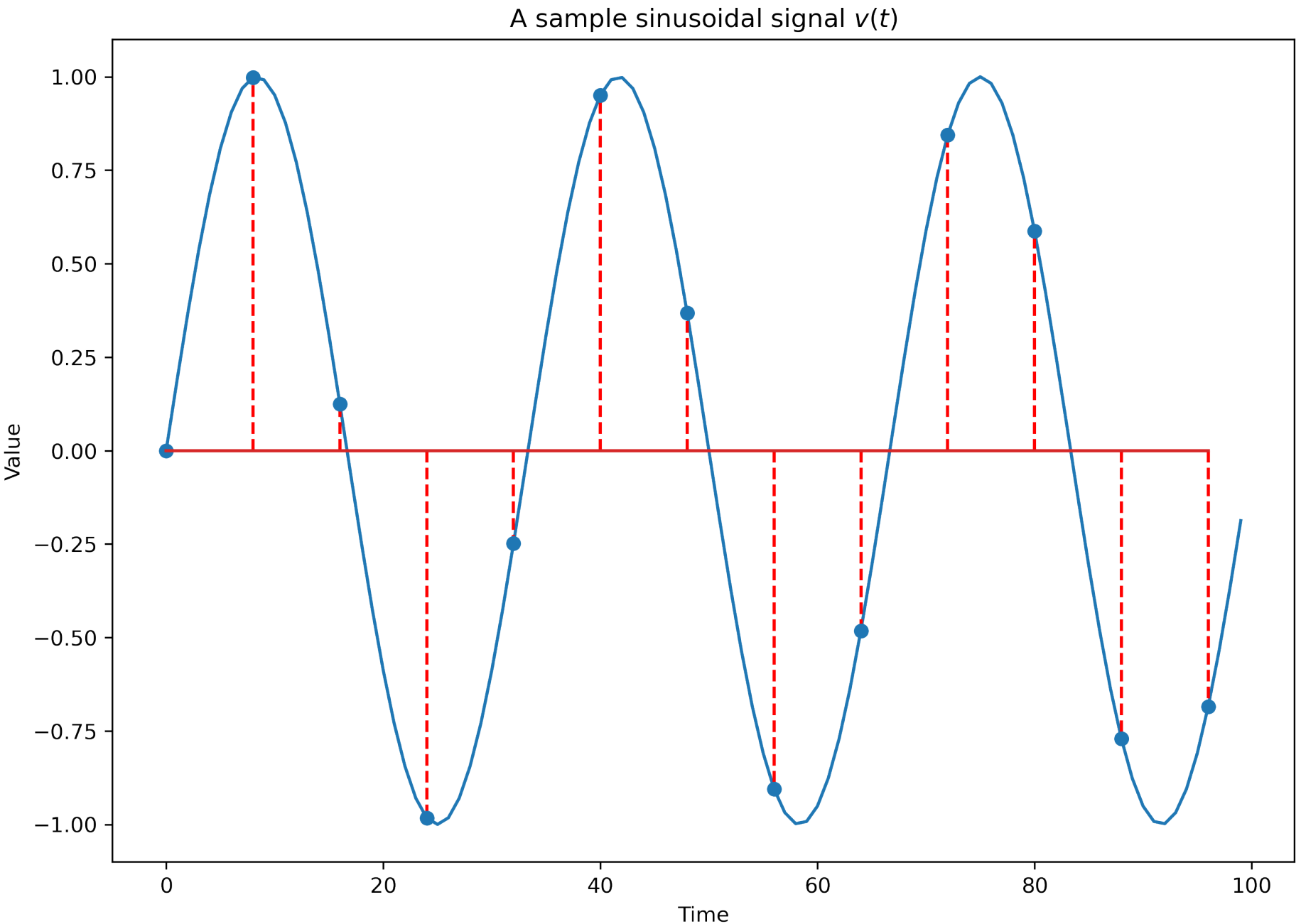
- Sampling = Taking the values from an analog signal at certain discrete moments of time, usually periodic
- Distance between two samples = **sampling period** T_s
- **Sampling frequency** $F_s = \frac{1}{T_s}$
- Why sampling?
  - Converts continuous signals to discrete
  - Processing of continuous signals is expensive
  - Processing of discrete signals is cheap (digital devices)
  - Sometimes nothing is lost due to sampling

From the physical world to digital processing



- ADC: **sampling** → **quantization** → **coding**
- Filters act in continuous time, before the ADC and after the DAC

Graphical example



Sampling equation

- Mathematically, it is described by **the sampling equation**, which uses the Dirac delta function $\delta(t)$:

$$x[n] = \int_{-\infty}^{+\infty} x_a(t)\delta(t - nT_s)dt$$

- For continuous signals, this is equal to the value of the signal at $t = nT_s$:

$$x[n] = x_a(n \cdot T_s)$$

- Periodic sampling produces a discrete signal $x[n]$ from a continuous signal $x_a(t)$

Sampling of harmonic signals

- Let's sample a cosine: $x_a(t) = \cos(2\pi F t)$

$$
\begin{aligned}
x[n] &= x_a(nT_s) \\
&= \cos(2\pi F n T_s) \\
&= \cos(2\pi F n \frac{1}{F_s}) \\
&= \cos(2\pi \underbrace{\frac{F}{F_s}}_{f} n) \\
&= \cos(2\pi f n)
\end{aligned}
$$

- Same for sine instead of cosine

Discrete frequency is relative

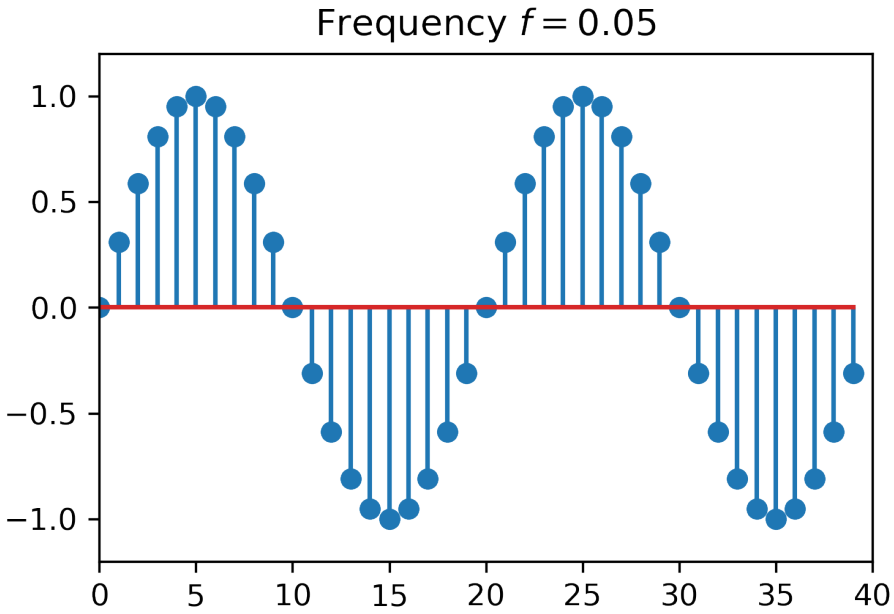
- Sampling a continuous cosine produces a discrete cosine with **discrete frequency**:

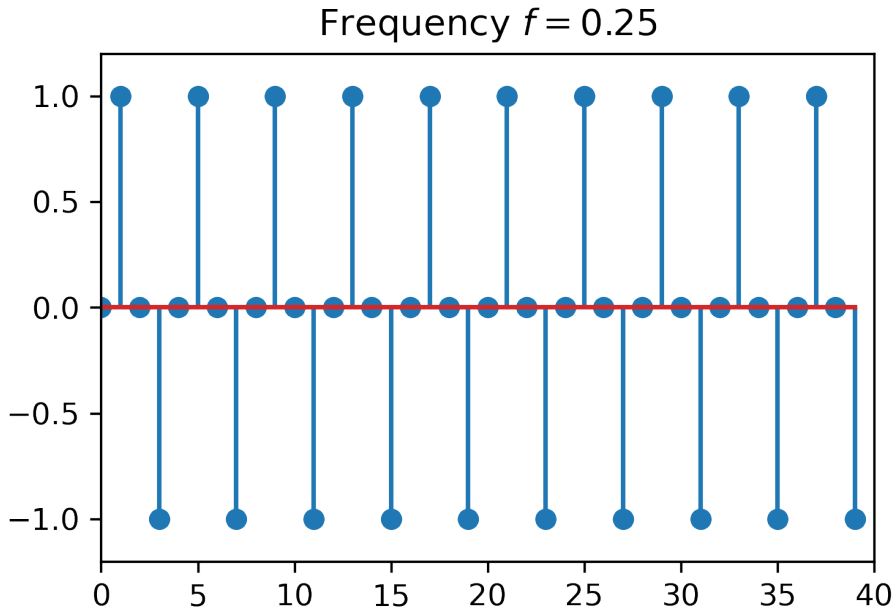
$$f = \frac{F}{F_s}$$

- Discrete frequency should be understood as a value **relative to the sampling frequency**
- Example: $f = \frac{1}{4}$ means "coming from an analog frequency F which was $\frac{1}{4}$ of the sampling frequency"
  - it could have been a 100Hz signal sampled with 400Hz
  - it could also have been a 3MHz signal sampled with 12MHz

False friends

- **Note:** A discrete sinusoidal signal might not *look* sinusoidal, when its frequency is high (close to $\frac{1}{2}$).





Sampling theorem (Nyquist-Shannon)

The Nyquist-Shannon sampling theorem:

- If a signal $x_a(t)$, which is band-limited with maximum frequency F_{max}, is sampled with a sampling frequency

$$F_s > 2F_{max},$$

then it can be perfectly reconstructed from its samples using the formula:

$$x_a(t) = \sum_{n=-\infty}^{+\infty} x[n] \cdot \frac{\sin(\pi(F_s t - n))}{\pi(F_s t - n)}.$$

Comments on the sampling theorem

- All the information in the original signal is contained in the samples, provided the sampling frequency is high enough
- It is much easier to process discrete samples instead of analog signals (e.g. using Matlab instead of capacitors :))
- Sampling with $F_s > 2F_{max}$ makes every signal frequency strictly smaller than $1/2$:

$$f = \frac{F}{F_s} \leq \frac{F_{max}}{F_s} < \frac{1}{2}$$

Example of the sampling theorem in action

Sampling theorem in action:

- Humans can only hear sounds up to ~20kHz
- Sampling rates higher than 40kHz are enough for audible frequencies
  - Standardized for CD-Audio: 44100Hz

Aliasing

- http://www.dictionary.com/browse/alias:
  - "alias": a false name used to conceal one's identity; an assumed name
- What happens when the sampling frequency is not high enough?
- Example: $F = 600Hz$ sampled with $F_s = 1000Hz$

$$\begin{aligned} x[n] &= x_a(nT_s) \\ &= \cos(2\pi 600 nT_s) \\ &= \cos(2\pi 600 n \frac{1}{1000}) \\ &= \cos(2\pi \underbrace{\frac{6}{10}}_{f} n) \end{aligned}$$

- Bad sign: We get a frequency larger than $f_{max} = \frac{1}{2}$

Funny things with cos() and sin()

- Discrete cos() and sin() have funny properties
- They are **the same** when adding an integer to the frequency:

$$\cos(2\pi(f+k)n) = \cos(2\pi fn + 2\pi kn) = \cos(2\pi fn)$$

- So all these discrete frequencies are identical:

$$f = ... = -1.4 = -0.4 = 0.6 = 1.6 = 2.6 = 3.6 = ...$$

- In addition, negative frequencies can be turned into positive:

$$\cos(2\pi(-f)n) = \cos(2\pi fn)$$

$$\sin(2\pi(-f)n) = -\sin(2\pi fn)$$

Aliasing

Aliasing:

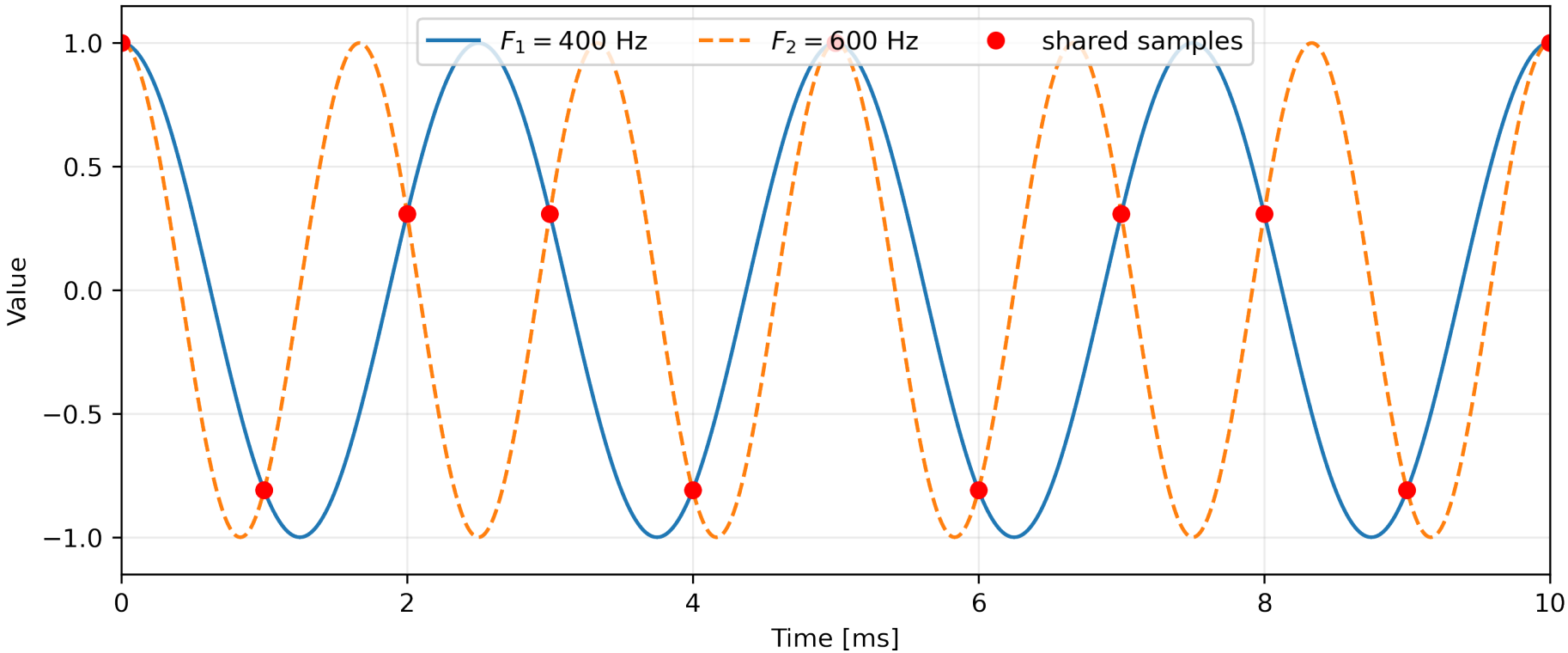
- Every discrete frequency f outside the interval $[-\frac{1}{2}, \frac{1}{2})$ is **identical** (an "alias") with a frequency from this interval $f_{alias} \in [-\frac{1}{2}, \frac{1}{2})$
- Just add or subtract 1's to f until the result is in $[-\frac{1}{2}, \frac{1}{2})$
- The interval $[-\frac{1}{2}, \frac{1}{2})$ is called the **Nyquist interval** or the **principal interval**

Frequency limits

- For continuous signals, F can go to ∞
  - Because period T can be $T \to 0$
- For discrete signals, we consider the **largest frequency** to be $f_{max} = \frac{1}{2}$, in the sense that all frequencies f can be reduced to a frequency $\in [-\frac{1}{2}, \frac{1}{2})$

One set of samples, two analog frequencies

- Let $F_s = 1000$ Hz, $F_1 = 400$ Hz and $F_2 = 600$ Hz
- Because $0.6 \equiv -0.4 \pmod 1$, the two cosines have the same samples



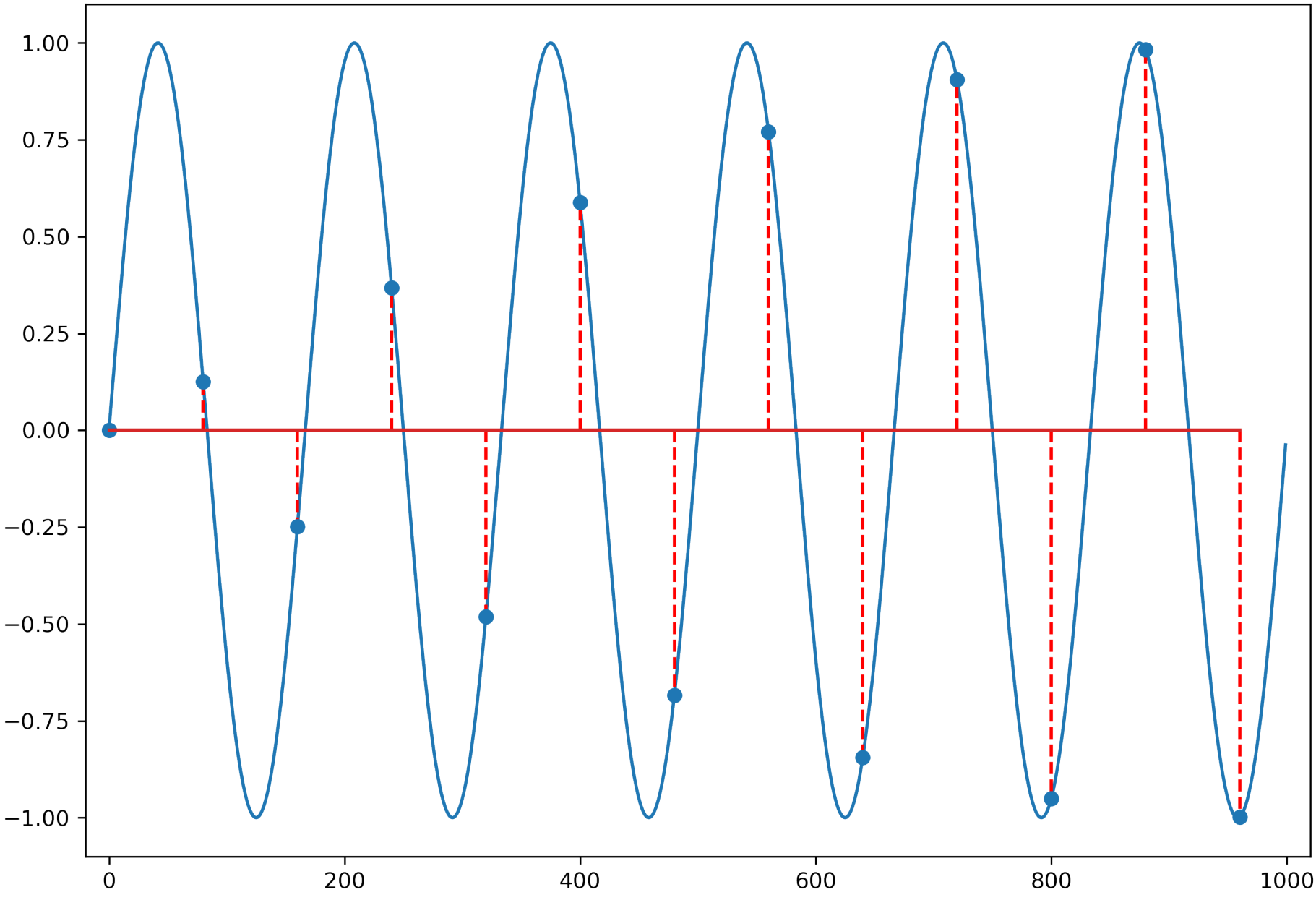
The problem of aliasing

- Sampling different signals can lead to exactly same samples
- Problem: how to know from what signal did the samples come from? Impossible.
- Example:
  - all these discrete frequencies are identical:

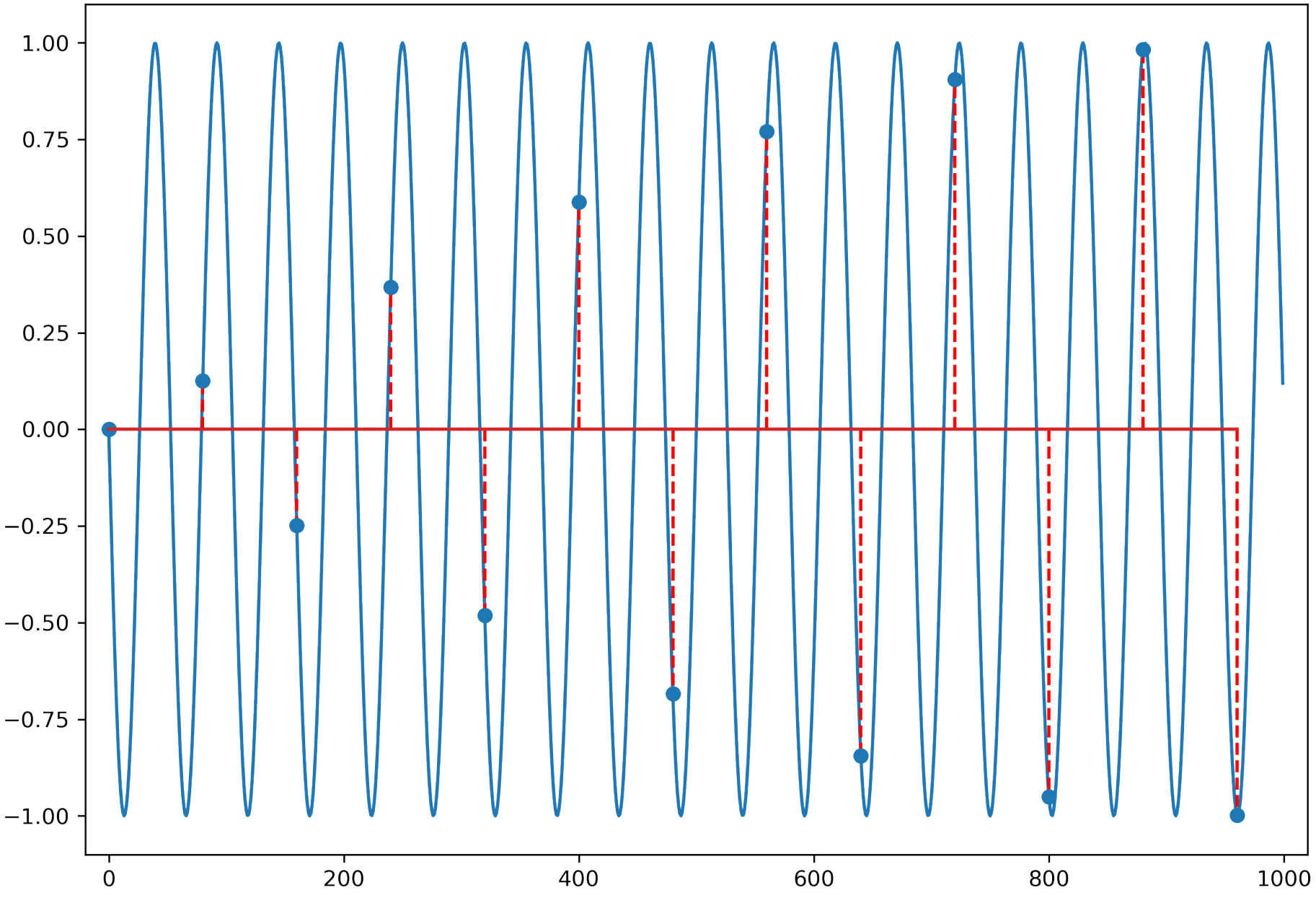
$$f = -0.4 = 0.6 = 1.6 = ...$$

  - so if $F_s = 1000Hz$, the original signal could have been any frequency F out of: 600Hz or 1600Hz or ...
  - Exercise: check some of these

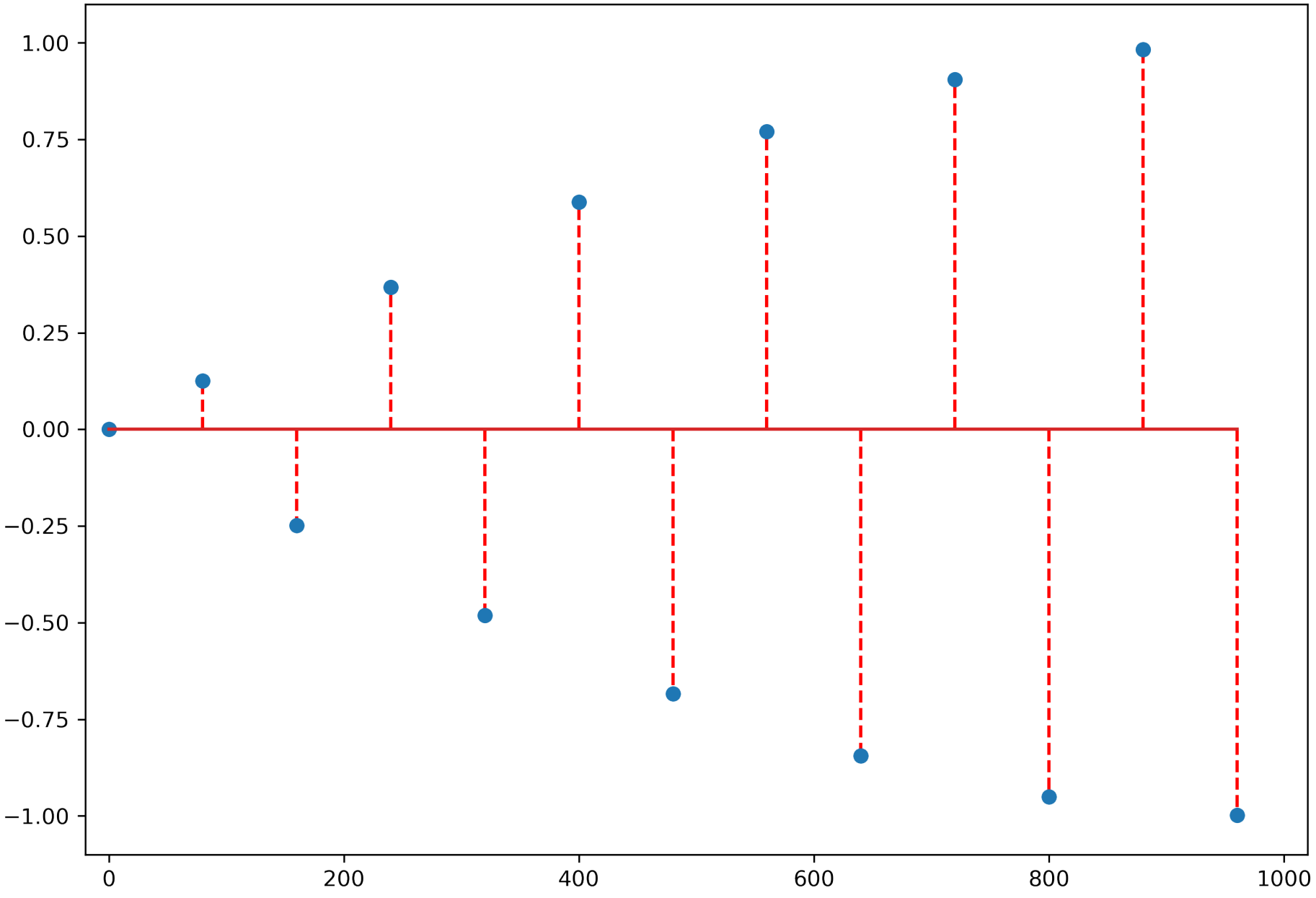
Aliasing example - low frequency signal



Aliasing example - high frequency signal, same samples



Aliasing example - samples only



Checkpoint: sampling and aliasing

- A 800 Hz cosine is sampled at 1000 Hz. Which cosine in the principal interval has the same samples?

Anti-alias

- Sampling according to Shannon theorem guarantees no aliasing:

$$F_s > 2F_{max} \Rightarrow f = \frac{F}{F_s} < \frac{1}{2}$$

- It is better remove from the signal all the frequencies larger than $\frac{F_s}{2}$, which will not be sampled correctly, otherwise they will create a false frequency and bring confusion
- **Anti-alias filter**: a low-pass filter situated before a sampling circuit, rejecting all frequencies $F \geq \frac{F_s}{2}$ from the signal before sampling
  - Standard practice in the design of processing systems

Ideal signal reconstruction from samples

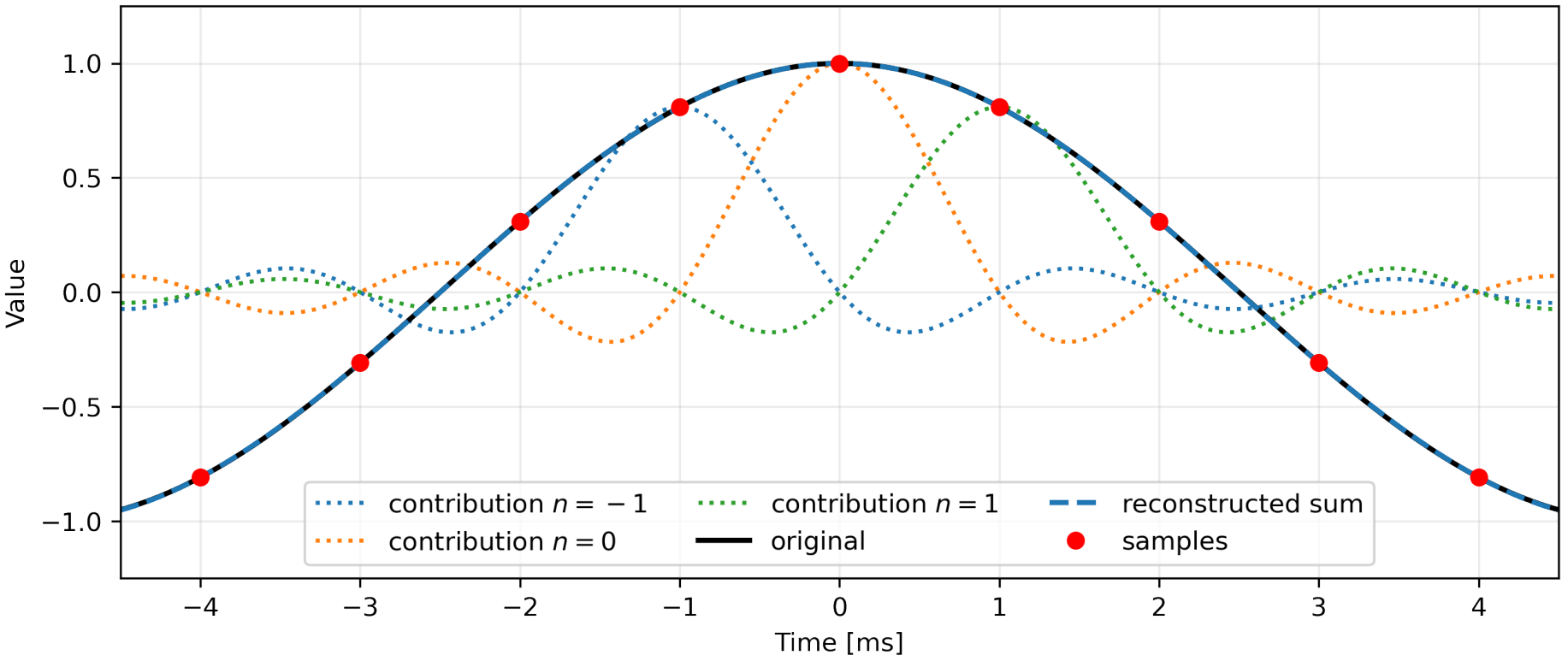
- Reconstruction = opposite of sampling, producing a continuous signal from a discrete one
- For harmonic signals ($\sin$, $\cos$), the ideal reconstruction equation amounts with the inverse variable change $n = \frac{t}{T_s} = t \cdot F_s$:

$$x_r(t) = x[\frac{t}{T_s}] = x[t \cdot F_s]$$

- A discrete frequency f becomes $F = f \cdot F_s$

Sinc interpolation: ideal reconstruction

- Example: $F = 100$ Hz sampled at $F_s = 1000$ Hz, safely inside the Nyquist interval
- We use 61 samples and show only the interior; shifted sinc terms sum through every sample value



Reconstruction and aliasing

- What value to use for f?
  - we know $f = f + 1 = f + 2 = ...$, which one to use?
- The reconstruction assumes all f are in the interval $[-\frac{1}{2}, \frac{1}{2})$
  - apply reconstruction equation
  - the resulting signal has all frequencies $\frac{F_s}{2} \leq F < \frac{F_s}{2}$ ($\frac{F_s}{2}$ = "the Nyquist frequency")
- **In exercises:** Always bring f in the interval $[-\frac{1}{2}, \frac{1}{2})$ before reconstruction
- Reconstruction always produces signals with frequencies in $[-\frac{F_s}{2}, \frac{F_s}{2})$
  - Only signals or components sampled according to the sampling theorem will be reconstructed identically
  - Any other components are replaced with their aliased counterparts

A/D and D/A conversion

- Sampling + quantization + coding is usually done by an **Analog to Digital Converter (ADC)**
  - It takes an analog signal and produces a sequence of binary-coded values
- Reconstructing an analog signal from numeric samples is done by a **Digital to Analog Converter (DAC)**
  - Usually the reconstruction is not based on sampling theorem equation, which is too complicated, but with simpler empirical solutions
- You have ADCs and DACs for any In or Out audio jack (phone, computer etc)

Signal quantization and coding

- In practice, the amplitudes of the samples are converted to binary representation
- Because of this, the amplitudes are rounded to fixed levels, e.g. 8-bit values (256 distinct levels) , 16-bit values (65536).
- This “rounding” is known as **quantization**
- The “rounding error” is known as **quantization error**
- Converting the value to binary form is known as **coding**
- ADCs handle sampling, quantization and coding simultaneously

Sampling: formula summary

- Sampling clock: $F_s = \frac{1}{T_s}$
- Sampling equation: $x[n] = x_a(nT_s)$
- Relative frequency: $f = \frac{F}{F_s}$
- Periodicity: $f = \frac{p}{q}$ in lowest terms $\iff N = q$; irrational f is not periodic
- Nyquist–Shannon condition: $F_s > 2F_{max}$
- Frequency equivalence: $f \equiv f + k \pmod 1$, $k \in \mathbb{Z}$; use $f \in [-\frac{1}{2}, \frac{1}{2})$
- Sinc interpolation:

$$x_a(t) = \sum_{n=-\infty}^{+\infty} x[n] \, \frac{\sin(\pi(F_s t - n))}{\pi(F_s t - n)}$$

- B-bit quantization provides 2^B amplitude levels